JaleesWeights: Internalizing the Righteous-Companion Disposition into Open Weights

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Abstract

JaleesBench, this paper's companion, measured whether AI assistants are good spiritual company for people of faith and found a consistent gap: every frontier model has a large *guided* ceiling — a one-page companionship guide in context makes it a genuinely good companion — but out of the box, with the guide absent, most models are middling and some are actively harmful. Real deployment rarely grants the prompt. We ask whether the disposition can instead live in the weights. We present a two-stage recipe — *judge-filtered context distillation* of the model's own guided outputs, followed by *on-policy preference sharpening anchored at the distilled checkpoint* — that takes an open 31B model (Gemma-4-31B) from the benchmark pool's bottom tier (−0.34 post-pressure) past the best open base model (+0.50 vs. +0.40) with the guide nowhere in context, while *raising* the model's guided ceiling (+0.57 → +0.80): weights and context compose. Preference optimization alone cannot do this: five DPO arms across two models and three data constructions all land statistically at zero, because contrast objectives cannot create behavior the base policy barely samples — in 317 of 420 training cells, four high-temperature draws yield no example worth reinforcing. Distillation is what unlocks preference learning: it flips the policy's own sample distribution from majority-cave to majority-hold, tripling pair-mining yield, and the preference stage then chiefly buys *steadfastness under pressure* (paired per-cell gain +0.22, 95% CI [+0.14, +0.30], nearly eliminating the under-pressure drop) — a gain that reproduces across two independent end-to-end runs of the pipeline (+0.220 under QLoRA, +0.223 under bf16). The distillation stage replicates on a second model family (Inkling-Small, 266B): bare post-pressure +0.25 → +0.55 (paired per-cell +0.30), the guided ceiling again rises, and the under-pressure drop is eliminated outright — leaving the preference stage, run identically, nothing to buy (paired +0.02): sharpening's value tracks the residual drop distillation leaves. A capability panel (MMLU, GSM8K, IFEval) shows no regression at either stage. We also contribute the evaluation discipline these claims required: same-serving-stack controls (the stack alone shifted scores by more than several apparent effects), a measured ∼29% per-resample cell-flip floor, and paired per-cell tests.

1 Introduction

JaleesBench [Kadous and Olsen, 2026] measures whether an AI assistant is a *righteous companion* (jalīs ṣāliḥ) — judged not by what the model knows or professes but by the residue its counsel leaves on the person who receives it. Its most actionable finding was about *prompting*: a one-page companionship guide lifts every frontier model from middling (+0.23–0.28) to genuinely good (+0.84–0.91), on par with a domain-tuned assistant. The disposition, in other words, is *prompt-reachable* in every model tested.

But deployment rarely grants the prompt. The person who opens a chat app and brings it a real decision does not paste a companionship guide first; the browser extension, the local model on a laptop, the API-mediated integration — none of them carry it. And the models most likely to be run in settings a Muslim family controls — open-weight models run locally — are precisely the ones that scored at the bottom of the pool bare. The gap between the guided ceiling and the bare score is therefore not a curiosity; it is the difference between what these systems could be for their users and what they are.

This paper asks whether that gap can be closed in the weights: can an open model be fine-tuned so that the righteous-companion disposition is simply *how it behaves*, with no guide in context? We answer yes, with a recipe whose structure is itself a finding. Our contributions:

1. **A two-stage internalization recipe.** Stage 1 distills the model's own guided outputs — filtered by a selection judge, re-rendered with the guide removed — into the weights by supervised fine-tuning; stage 2 sharpens the distilled policy with on-policy preference pairs mined from its own samples, using the distilled checkpoint as the DPO reference. On Gemma-4-31B the pipeline moves post-pressure performance from -0.335 to $+0.499$ — past the best open base model in the pool — and raises the guided ceiling from $+0.569$ to $+0.802$, for under \$100 of compute (sections 5 and 6).
2. **A mechanism account of why preference optimization alone fails.** Five DPO arms on the base policy — two models, three data constructions, doses from 179 to 1,200 pairs, training accuracies up to 1.0 — all land statistically at zero. The binding constraint is *samplability*: in 317 of 420 training cells the base policy never samples an example worth reinforcing, and a contrast objective cannot create behavior the policy cannot produce (section 4). Distillation is what makes preference learning possible: it moves the sample distribution first (section 6.3).
3. **Evaluation discipline for single-sample-per-cell benchmarks.** Chasing spurious "degradations" taught us that the serving stack alone shifts scores by more than several apparent effects, and that any two collections disagree on $\sim$29% of cells. We describe the controls that make small-effect claims on such benchmarks trustworthy: same-stack baselines, paired per-cell tests, and aggregate confidence intervals over per-cell readings (section 4.4).

Everything runs on a single GPU with LoRA; data, drivers, and judgments are open-sourced with the benchmark. The result generalizes the Constitutional AI observation [Bai et al., 2022] — a supervised stage that moves the distribution, then a preference stage that sharpens it — to a setting where the target is not harmlessness but a positive, thickly-specified relational disposition, and where the teacher is the model's own guided behavior rather than a separate critique-revision process.

2 Related work

Three strands of prior work meet in this paper: context distillation as a way of moving prompts into weights, preference optimization and its known failure modes, and the small literature on tuning models toward normative dispositions.

2.1 Context distillation and constitutional pipelines

Askell et al. [2021] introduced *context distillation*: fine-tune a model on its own outputs generated with a prompt in context, with the prompt removed from the training input, so that the prompt's effect migrates into the weights. Constitutional AI [Bai et al., 2022] builds a two-phase pipeline in this family: a supervised phase in which the model critiques and revises its own outputs against a set of principles and is fine-tuned on the revisions, followed by a reinforcement phase (RLAIF) driven by

AI-generated preference labels. Our stage 1 is the direct-distillation cousin of their supervised phase: we skip critique-revision entirely and instead *filter* — the teacher set is the model's own guided outputs that an independent selection judge scored as genuinely good companionship, re-rendered without the guide. Our stage 2 occupies the structural position of their reinforcement phase. The composition finding of section 6.2 — distilled weights plus the original prompt outperform either alone — matches the spirit of Askell et al.'s observation that distillation and prompting are complementary rather than redundant.

2.2 Preference optimization and its limits

Direct Preference Optimization [Rafailov et al., 2023] collapses RLHF's reward-model-plus-PPO loop into a single contrastive loss over preference pairs, and has become the workhorse of open-model alignment [Tunstall et al., 2023]. A recurring theme in subsequent work is that *where the pairs come from matters*: methods that sample from the current policy and re-label — rejection-sampling fine-tuning [Dong et al., 2023], ReST and ReST-EM [Gulcehre et al., 2023, Singh et al., 2024], self-play and self-rewarding schemes [Chen et al., 2024b, Yuan et al., 2024] — consistently outperform training on static, off-policy pairs, and Tajwar et al. [2024] argue on-policy data is the key variable. A second theme is that the DPO objective optimizes a *ranking*, which can decouple from generation: trained models often fail to rank their own outputs consistently with the training preference [Chen et al., 2024a], and the loss can be satisfied by displacing probability mass away from both sides of the pair [Pal et al., 2024, Razin et al., 2024]. Our negative results sharpen both themes into a single mechanism: on a behavior the base policy barely samples, *no* pair construction — off-policy, on-policy, or pooled relative-preference — moves held-out behavior, even with training accuracy saturated at 1.0 (section 4.2). The on-policy prescription is necessary but not sufficient; the policy must first be moved to where its own samples contain the target behavior, which is precisely what distillation does (section 6.3).

2.3 Value and faith-conditioned fine-tuning

Work at the intersection of language models and Islam has measured knowledge [Abdelaal et al., 2026, Elmahjub et al., 2026] and professed values [Lahmar et al., 2025]; JaleesBench measures benefit to the user, and this paper is (to our knowledge) the first attempt to *train* an open model toward that construct. The nearest antecedent is prompt-level: JaleesBench's own guide experiment showed the disposition fits in a page of instruction. Constitutional AI [Bai et al., 2022] tunes toward a trait catalogue (harmlessness); our target differs in being a positive, relational disposition — warmth that survives pressure, counsel anchored in a tradition's own sources — whose ground truth is a classical virtue compilation rather than a list of prohibitions. The recipe is faith-general in the same sense the benchmark is: nothing in it is specific to Islam except the evaluation instrument and the guide being distilled.

3 Setup

We give the minimum instrument recap needed to read the results, then the protocol decisions specific to this paper: the task definition, the data discipline, and the training envelope. Full instrument details — scenario construction from Riyāḍ al-Ṣāliḥīn, pressure taxonomy, judging rubric and its validation — are in the companion paper [Kadous and Olsen, 2026].

JaleesBench evaluates a system in *sittings*: two-turn conversations in which a user brings a real decision, receives counsel, then pushes back with one of six adversarial *pressures* (insistence, flattery, personal appeal, false authority, appeal to a good cause, secularization). Each of 140 scenarios runs under three *framings*; this paper works in the *unstated* framing — the user's faith is inferable but never announced — because it is the deployment condition. Judges score the counsel's effect on the user on a five-band scale (native $-2 \dots +2$, reported normalized to $-1 \dots +1$) at two scopes: the first response (turn 1) and the response after pressure (post-pressure). Post-pressure bare — no guide, faith unstated, after the user pushes back — is this paper's target metric, and the hardest condition in the benchmark.

3.1 The internalization task

The companion paper's guide experiment defines the teacher signal. With a one-page companionship guide (GUIDE_MIN) in context, Gemma-4-31B scores +0.57 post-pressure; bare it scores −0.34 — the pool's bottom tier. The internalization task: fine-tune the model so its *bare* behavior approaches its own *guided* behavior, using only artifacts the benchmark already produces (the model's guided sittings and the judges' bands). Success is measured on held-out scenarios the model never trained on, bare, after pressure.

3.2 Split, judge holdout, and controls

Three disciplines govern every experiment in this paper.

Scenario split. The 140 scenarios are split 70/70 (seed 3446), stratified by register and primary pillar; all training data — SFT examples and preference pairs alike — is mined from train-70 sittings only, and every reported number is on test-70. Splits are by scenario, never by judgment, so no scenario contributes to both sides.

Judge holdout. Two frontier judges score the main benchmark; we assign them non-overlapping roles. Gemini selects all training data (SFT filtering, pair selection); Opus judges all held-out evaluations and never touches selection. A model tuned on one judge's preferences is thus evaluated by a judge that had no hand in its training data. Because this paper's evaluations are Opus-only on test-70 (the companion paper's headline numbers are two-judge over all 140 scenarios, and Opus scores warmer), baselines are re-pinned under this paper's protocol: Inkling +0.398 post-pressure, Gemma-4-31B −0.276 as originally served.

Same-stack controls. Midway through the study we found that serving infrastructure alone moves scores: base Gemma evaluated through our vLLM harness scores −0.335 post-pressure vs. −0.276 from the provider that served the main run — a −0.058 shift, larger than several arms' apparent effects, on identical weights (section 4.4). Every comparison in this paper is therefore against a base-model control collected through the *identical* serving path as the tuned checkpoint: −0.335 post-pressure, −0.044 turn 1. Uncertainty is reported as 95% scenario-cluster bootstrap intervals, and checkpoint-vs-checkpoint claims additionally use paired per-cell comparisons on identical scenario–pressure cells.

3.3 Models and training envelope

The subject model is Gemma-4-31B (open weights, Apache 2.0) — chosen deliberately from the pool's bottom tier: if internalization works where the bare disposition is weakest, the headroom claim is strongest. Training of record is bf16 LoRA [Hu et al., 2022] — rank 32 on the language-model projection matrices — on a single B200; the negative arms of section 4 and an earlier end-to-end run of the pipeline used QLoRA [Dettmers et al., 2023] (4-bit NF4 base, same rank) on an H200, and section 6.6 reports the two pipeline runs against each other. Training examples are sitting-level: both assistant turns contribute to the loss, user turns (including the authored pressure turn) are in context but loss-masked, matching the chat template used at collection exactly. A startup parity check verifies the training-side forward pass reproduces the model's own logits before any step (this caught Gemma-4's final-logit softcapping, which would otherwise have silently corrupted DPO's log-probability ratios). The full two-stage pipeline — sampling, selection, both training stages, evaluations, and guards — cost under $100 for this model. The replication model is Inkling-Small (266B, open weights), trained as a LoRA through the Tinker fine-tuning API at the same stage-1 dose and evaluated through the same collection driver as its own baselines (section 6.7).

4 Preference optimization alone does not move the disposition

The obvious first move is DPO on the benchmark's own data: the main run already contains, for most training cells, a good response and a bad one — a preference pair with no new generation cost. We ran five arms of this family — two models, three data constructions, doses from 179 to 1,200 pairs — before the recipe of section 5 emerged, and every arm landed statistically at zero (Table 1). This section is the mechanism account: why an objective that demonstrably learned its training preferences moved nothing, and why the failures were invisible until the measurement discipline of section 3.2 was in place. Readers who want the recipe can skip to section 5; the boundary mapped here is what motivates its structure.

4.1 The samplability boundary

The decisive diagnostic was not a training curve but a sampling census. We drew $K = 4$ independent chains per train-70 cell (420 cells, 1,680 sittings) from base Gemma at elevated temperature and had the selection judge band them all. In **317 of 420 cells — 75% — no draw reached the good-band threshold** (band $\geq +1$): there was nothing to reinforce. 42 cells held in every draw (nothing to contrast against), leaving 58 cells — 14% — with genuine within-policy contrast. An on-policy preference method can only reinforce behavior the policy itself samples; for this disposition, under pressure, the base policy barely samples it at all. Off-policy pairs evade the boundary only nominally: the "chosen" behavior is then another model's text, and Table 1 shows what that buys. **[TODO: samplability histogram figure from judgments_train_samples.jsonl]**

4.2 Ranking is not behavior

Every arm learned its training task. Preference accuracy ranged from 0.61 (Inkling, off-policy, the shannon-calibrated conservative dose) to complete saturation at 1.00 (Gemma, off-policy: margins exploded, loss ≈ 0) — and the saturated arm was exactly as flat as the under-trained one. This rules out dose as the binding constraint. It also localizes the failure: when chosen completions are predominantly *other* models' text, the objective teaches the policy to *rank* alien styles above its own cave-in — a discrimination task it can ace without its own generations under pressure moving at all. The on-policy arm (all pairs the model's own text, the construction the literature prescribes; section 2.2) was the strongest-designed test and is the most instructive zero: confined by the samplability boundary to 58 cells, it did not move behavior even in the pressure category where its training coverage was densest (false authority, 16 of 58 cells: $\Delta + 0.022$, noise).

Arm	Model	Pairs (cells)	Chosen text	Pref. acc.	Δ post vs. control
r1 off-policy	Inkling	254	other models'	0.61	−0.025
r2 off-policy, 4.7×	Inkling	1,200	other models'	0.66	−0.013
r1 off-policy	Gemma	249	other models'	1.00	−0.015
on-policy	Gemma	179 (58)	own	0.83	+0.023
pooled max-gap	Gemma	516 (132)	pooled, relative	0.87	−0.005
stage 1 distillation (section 5.1)	Gemma	—	own guided	—	**+0.523**

Table 1: Five DPO arms on the base policy, all statistically zero against the same-stack control (test-70, unstated, post-pressure, Opus-judged; scenario-cluster bootstrap CIs overlap zero throughout). Training preference accuracy rises to complete saturation with no effect on held-out behavior. One pass of supervised distillation on the same model, from the same benchmark data, moved it +0.52 (QLoRA run; the bf16 run of record moves it +0.61, Table 2).

4.3 Temperature does not rescue sampling

The sampling census ran hot (temperature 1.3 vs. the model's configured default) precisely to give the policy its best chance of surfacing rare good behavior. It did not: hot chains band-distribute essentially identically to the model's default-temperature main run on the same cells ($\{-2: 59\%,\ 0: 20\%,\ \geq +1: 17\%\}$ hot vs. $\{-2: 56\%,\ 0: 23\%,\ \geq +1: 17\%\}$ default). Cave-versus-hold under pressure is *scenario-determined*, not sampling luck; heat neither rescues hard cells nor degrades easy ones. The bump's real value was as a mining tool: four draws per cell surface within-cell contrast in the minority of cells where the model's behavior is genuinely bistable.

4.4 What looked like harm was measurement

Early readings suggested the DPO arms made the model slightly *worse* (−0.276 → −0.349 and −0.311), and per-cell inspection found dramatic individual collapses. Both readings dissolved under controls, and the dissolution is itself a methodological contribution for anyone evaluating fine-tunes on single-sample-per-cell benchmarks.

First, the apparent degradation was the serving stack. Base Gemma — no adapter, identical weights — run through the same vLLM evaluation harness as the tuned checkpoints scores −0.335, not the −0.276 the main run's provider produced. Against the correct same-stack control, the arms' deltas are −0.015, +0.023, −0.005: noise (Table 1).

Second, the dramatic per-cell collapses were resampling variance. The no-training control itself flips 122 of 420 cells against the original baseline collection — a ∼29% cell-flip floor between *any* two collections — including 22 collapses of two bands or more on identical weights. The arms' collapse lists are largely the control's: 15 of 24 (r1) and 16 of 20 (on-policy) of their big collapses occur in the same cells the control collapses in, and across the two independently-trained arms, 71 of 79 co-moving cells moved in the same direction — movement that cannot be training-specific. Transcripts show the mechanism: the model is bistable per cell. On the same flattery-pressure scenario, pure base Gemma in one draw refuses firmly ("I will not give you the permission you are looking for") and in another caves warmly ("Yes, it is absolutely fine to let this specific routine go"). A cell "getting worse" usually means landing on cave-mode this draw.

The consequence we adopt throughout: on this class of benchmark, per-cell claims require a paired same-stack control, and aggregate cluster-bootstrap intervals are the only trustworthy readout of an arm's effect.

5 The JaleesWeights recipe

The two stages do jobs the other cannot. Distillation *creates* the behavior — it is a supervised objective, so it can move the policy toward text the policy would never have sampled — but it imitates the teacher's average, pressure-yielding included. Preference contrast cannot create, but once the distilled policy samples the target behavior richly, it can *consolidate*: push the policy's own good modes over its own bad ones under pressure. Stage 1 clears the samplability boundary; stage 2 cashes it.

5.1 Stage 1: judge-filtered context distillation

The teacher set is the model's own guided sittings from the main run, selected and screened: Gemini band $\geq +1$ at *both* scopes (turn 1 and post-pressure), a guide-reference screen (no response may mention or paraphrase the guide's existence; zero hits), and a dangling-citation screen (no bracketed reference markers that resolve nowhere; zero hits) — 316 sitting-level examples. Because the guide lives in a context prefix outside the conversation turns, re-rendering the sitting *without* it is exact: the training input is precisely what a bare deployment would see, completed by what the model itself said when guided. This is context distillation in the Askell et al. [2021] sense with a judge-filter in place of critique-revision. Training: masked NLL on both assistant turns (bf16 LoRA r32, lr 5×10^{-5}, 2 epochs, batch 8, 79 steps; per-token NLL $2.6 \rightarrow 0.49$).

5.2 Stage 2: on-policy sharpening at the distilled anchor

Stage 2 repeats the sampling census of section 4.1 — $K = 4$ chains per train-70 cell — but from the *distilled* policy, whose sample distribution has moved (section 6.3). Gemini bands all samples; pairs are mined max-gap, both directions: each sample is anchored against its most-differently-banded within-cell counterpart at a native gap ≥ 2, so the objective sees relative preference among the policy's own outputs rather than an absolute quality threshold — 502 pairs over 165 cells. Training is DPO with the *SFT checkpoint as reference* — the reinforcement-phase position in a constitutional pipeline [Bai et al., 2022], implemented as two adapters (policy initialized from and verified identical to the reference at step 0) — β 0.1, lr 1×10^{-5}, 1 epoch, batch 8, 63 steps, final preference accuracy 0.99.

6 Results

Table 2 is the paper in one table: each stage's held-out scores against the same-stack control, with the best open base model and the recipe's own raised guided ceiling as reference lines. Each subsection below states one finding.

6.1 Distillation creates the behavior

One pass of judge-filtered context distillation moves post-pressure performance from -0.335 to $+0.276$ — $+0.61$ with non-overlapping intervals, after five preference arms had moved nothing (Table 1). At turn 1 the distilled model scores $+0.429$, statistically at the level of the pool's best open base model (Inkling, $+0.468$): before pressure is applied, a 31B model that was near the bottom of the pool now counsels at open-frontier tier. What distillation does *not* fix is the yield

Checkpoint	Turn 1	Post-pressure	Steadfastness (Δ)
Gemma base (same-stack control)	−0.044	−0.335	−0.290
Stage 1: distilled (`sft-guided`)	+0.429	+0.276	−0.152
Stage 2: sharpened (`sft-dpo`)	**+0.535**	**+0.499**	**−0.036**
Ref: Inkling base (best open subject, bare)	+0.468	+0.398	−0.070
Ref: Stage 1 + guide in context (ceiling)	+0.829	+0.802	−0.026

Table 2: The pipeline on test-70, unstated framing, Opus-judged (840 judgments per row), −1 … +1. 95% scenario-cluster bootstrap CIs: stage 1 post-pressure [+0.12, +0.42]; stage 2 post-pressure [+0.35, +0.63]; control [−0.47, −0.20].

under pressure: the distilled model still drops −0.152 from turn 1 to post-pressure — it learned the guided teacher's first-response quality more thoroughly than its steadfastness, which is exactly the gap stage 2 targets.

6.2 Weights and context compose: the guided ceiling rises

The natural worry is that internalizing the guide damages the model for users who *do* paste it. The opposite happens: the distilled model with the guide in context scores +0.802 post-pressure vs. base with the guide at +0.569 — and since the vLLM stack measures $\sim$0.06 colder than the provider that produced the base-guided number, the composition gain is if anything understated. Under-pressure drop with the guide shrinks from −0.167 (base) to −0.026 (distilled). Weights and context compose: the model absorbed roughly the mid-band of what the guide provides, and re-applying the prompt stacks on top rather than colliding. Guide users get a better model, not a degraded one. This also maps stage 2's target: after distillation, the guide still adds +0.53 bare-to-guided — that is the remaining internalizable behavior.

6.3 Distillation makes the policy's own data minable

The stage-2 sampling census quantifies why the pipeline is ordered as it is. From the base policy, four draws per cell yielded within-cell contrast in 58 of 420 cells (section 4.1); from the distilled policy, the sample band mode flips outright — from 59% at band −2 (cave) to 44% at band +2 (hold) — and max-gap mining yields 502 pairs over 165 cells, roughly triple the cell coverage and pair count. This is the empirical content of the constitutional "supervised before reinforcement" ordering [Bai et al., 2022]: the supervised stage is what makes the policy's own output distribution rich enough for preference methods to have something to work with.

6.4 Preference sharpening buys steadfastness

Stage 2 moves post-pressure from +0.276 to +0.499. The paired per-cell comparison against stage 1 — same cells, same judge, same stack — is +0.223 (95% CI [+0.144, +0.304]; 120 cells up, 34 down): decisively positive where Table 1's base-policy arms were decisively null, with the same objective family, the same selection judge, and a nearly identical dose. What changed is where the pairs came from (the policy's own now-rich distribution) and what the reference was (the distilled anchor). The gain lands precisely where distillation left the gap: the under-pressure drop shrinks from −0.152 to −0.036 — statistically indistinguishable from the guided-in-context condition (−0.026) — and while turn 1 also moves (+0.106, 95% CI [+0.036, +0.177]), the post-pressure gain is twice as large: the preference stage chiefly bought *steadfastness*, which is the outcome the pair construction targets —

pairs contrast hold-vs.-cave under pressure, not first-response quality. The final checkpoint exceeds the best open base model in the pool bare (+0.499 vs. Inkling's +0.398) and stands roughly three-quarters of the way from the same-stack control to the model's own raised guided ceiling (+0.802).

6.5 Guards hold: no citation fabrication, register intact

The recipe's failure modes would be a model that fabricates the tradition's sources or one that trades warmth for rigidity. Neither appears. Across all 420 evaluation sittings of each tuned checkpoint: zero bracketed citation markers (the fabricated-reference style the training screens excluded; base also emits zero), and an unchanged length profile (median 5.8k characters at both stages vs. base 5.6k). Qualitatively, the cells where base Gemma was most dramatically bistable — the flattery and personal-appeal scenarios of section 4.4 — now hold in the warm companion register rather than by curt refusal.

6.6 The pipeline replicates across precision regimes

The numbers above are the bf16 run of record, but the pipeline was first run end-to-end under QLoRA (4-bit NF4 base, same rank and hyperparameters, H200) — an independent training run at every stage, including its own stage-2 sampling census and pair mining (518 pairs over 171 cells). The qualitative picture reproduces exactly: stage 1 +0.188, stage 2 +0.408, and a stage-2 paired per-cell gain of +0.220 (95% CI $[+0.129, +0.317]$) against the bf16 run's +0.223 — the pipeline's central effect, replicated on a second seed with independently mined training data. Paired per-cell comparisons of the two chains on identical cells give bf16 a small post-pressure edge at both stages (+0.088, 95% CI $[+0.010, +0.174]$ at stage 1; +0.090, $[+0.017, +0.164]$ at stage 2), with the paired turn-1 deltas null at both (exactly 0.000; +0.017, $[-0.051, +0.098]$). A plausible mechanism — consistent with the effect appearing only under pressure, though we do not claim it — is that the QLoRA chain trains against an NF4-quantized base but is served in bf16 by vLLM, and the bf16 chain removes that train/serve mismatch.

6.7 Distillation replicates on a second model family

Stage 1 replicates, and the effect is larger. Inkling-Small (266B, open weights, trained and served through the Tinker fine-tuning API) went through the identical stage-1 pipeline: its own guided train-70 sittings, judge-filtered under the same band and screens (310 examples), re-rendered bare, fine-tuned at the same dose (lr 5×10^{-5}, 2 epochs; NLL 1.34 → 1.15). Held out and bare, post-pressure moves +0.252 → +0.550 — paired per-cell +0.298 (95% CI $[+0.206, +0.392]$; 154 cells up, 37 down) — and turn 1 moves +0.394 → +0.524, on a model whose baseline was already competitive with the pool's frontier tier. The composition result replicates as well: the guided ceiling rises from +0.817 to +0.885. Most striking, distillation alone *eliminates* the under-pressure drop (−0.142 → +0.026; the tuned model scores at or above its turn-1 level after pushback): the steadfastness gap that stage 2 existed to close on Gemma does not survive stage 1 on this model — consistent with Inkling-Small's near-lossless guided baseline (drop −0.004 with the guide in context; the guide fully immunizes this model, and the immunity distilled along with everything else). Caveat: base and tuned checkpoints were collected through the same provider and driver, but the provider serves base weights and adapter weights through different lanes; the paired effect is ∼5× the largest serving-stack shift we have measured (section 4.4).

Stage 2, run identically on top (672 own-sample max-gap pairs from a $K{=}4$ census of the distilled policy, SFT-as-reference DPO at the same dose), is *flat* on this model: +0.569 vs. +0.550, paired

Benchmark	Base	Stage 1	Stage 2
MMLU (acc)	0.467	0.468	0.470
GSM8K CoT (strict)	0.792	0.867	0.848
IFEval (instruction-level strict)	0.273	0.279	0.282
IFEval (prompt-level strict)	0.164	0.155	0.157

Table 3: Capability panel, identical harness and serving configuration for all three checkpoints. No benchmark regresses; GSM8K improves at both tuned stages.

per-cell +0.019 (95% CI $[-0.032, +0.070]$), guards clean. We read the pair of outcomes as one mechanism, not a mixed result: the preference stage buys down the residual under-pressure drop, and its value tracks the size of that residual. On Gemma, distillation left a −0.152 drop and sharpening nearly eliminated it (−0.036; paired gain +0.223, and the same pattern in the QLoRA run: −0.243 left, +0.220 bought); on Inkling-Small, distillation left no drop (+0.026) and sharpening had nothing to buy (+0.019). The census had flagged the contingency in advance — the distilled policy still sampled cave-mode in 35% of train-cell draws under the selection judge — but that per-draw bimodality did not translate into evaluator-visible headroom on the held-out set. The portable recipe is therefore: distill always; then measure the residual drop, and sharpen only what remains.

6.8 General capability does not regress

A disposition bought at the price of general capability would be a bad trade for real deployment. We ran an lm-evaluation-harness [Gao et al., 2023] panel — MMLU [Hendrycks et al., 2021], GSM8K [Cobbe et al., 2021], IFEval [Zhou et al., 2023] — over all three checkpoints under an identical vLLM configuration (Table 3). MMLU and IFEval are flat within standard error across the pipeline; GSM8K *improves* at both tuned stages (+7.4 and +5.5 points over base). We read the GSM8K movement as distillation tightening chain-of-thought format compliance rather than as a reasoning gain, and do not claim it as a benefit; the load-bearing result is the absence of regression anywhere on the panel. The panel was run on the QLoRA chain's checkpoints (section 6.6); given the two chains' near-identical disposition results, we do not expect the bf16 checkpoints to differ.

7 Side effects: does the disposition over-apply?

The likeliest failure mode this paper has not yet measured is over-application: a model tuned to be a righteous companion for a Muslim user might volunteer religious counsel to users who did not ask for it, or let the disposition color unrelated assistant work. The capability panel (section 6.8) bounds the damage on standard tasks at zero, but instruction-following benchmarks do not probe conversational register. **[TODO: probe suite: neutral/secular prompts, non-Muslim interlocutors, uninvited-counsel rate — design pending; results decide whether this section reports a finding (the disposition stays context-appropriate) or a limitation]**

8 Limitations

Both stages have now run on two model families, but stage 2's *positive* result is one model deep: on the second family it was correctly-but-unfalsifiably flat, since distillation left it no residual drop to buy (section 6.7) — a third family with a Gemma-sized residual would test the sharpening claim properly. The Gemma pipeline has been trained end-to-end twice (the QLoRA run and the bf16 run of record), with the stage-2 paired gain reproducing (+0.220 and +0.223; section 6.6); every other arm and stage is one seed deep, and section 4.4's cell-flip floor is a reminder of how much single-collection readings can wobble (though the pipeline's headline effects are 5–10× that floor, and the paired test controls for it). Training is LoRA-only; whether full fine-tuning changes the picture — in either direction — is untested. Transfer is measured within the benchmark's own scenario distribution (held-out scenarios, but the same construction recipe and the same six pressures), and in English only; the companion paper's Arabic replication has not been run against the tuned checkpoints. The final model's steadfastness (−0.036) now matches the guided-in-context condition (−0.026), but its bare score sits at roughly three-quarters of its guided ceiling: internalization is substantial, not complete. Finally, both judges are frontier models from the same generation; the judge-holdout design keeps selection and evaluation independent, but a shared blind spot across both judges would be invisible to it.

9 Conclusion

The companion paper ended with a gap: the righteous-companion disposition is prompt-reachable in every frontier model, but the prompt is absent exactly where the stakes are real. This paper closes that gap in the weights for an open 31B model, and locates *why* the obvious approach fails: preference optimization cannot create behavior the policy barely samples, no matter how cleanly it learns to rank it. The recipe that works is ordered by that mechanism — distill the model's own judge-approved guided behavior to move the sample distribution, then sharpen with preference contrast among the policy's own outputs — and each stage's contribution is visible in the numbers: distillation buys the behavior (−0.335 → +0.276), sharpening buys the steadfastness (+0.276 → +0.499, drop −0.152 → −0.036), and the guided ceiling rises rather than falls (+0.569 → +0.802). The pipeline costs under $100 on one GPU, uses no data beyond what the benchmark itself produces, and generalizes in principle to any tradition the benchmark family instantiates: measure the disposition, guide it, then teach the model to be, bare, what the guide made it.

10 Future work

Four directions are queued. *Iteration*: stage 2's sampling census can be re-run from the sharpened checkpoint — does a second distill-sharpen round keep climbing toward the ceiling, and does judge gaming appear before it converges? *Proof-grounded distillation*: a variant whose teacher passes include the tradition's proof texts in the generation context, with hardened citation verification, testing whether internalized counsel can carry its sources honestly. *Critique-revision*: the constitutional supervised phase proper, compared head-to-head with our judge-filter. *Online RL*: the design reserve if preference-pair methods plateau. And the two measurement extensions the companion paper makes natural: the Arabic pipeline against these checkpoints, and confirmation on the full 140-scenario bank.

References

Ali Abdelaal, Mohammed Nader Al Haffar, Mahmoud Fawzi, and Walid Magdy. IslamicMMLU: A benchmark for evaluating LLMs on islamic knowledge, 2026.

Amanda Askell, Yuntao Bai, Anna Chen, Dawn Drain, Deep Ganguli, Tom Henighan, Andy Jones, Nicholas Joseph, Ben Mann, Nova DasSarma, Nelson Elhage, Zac Hatfield-Dodds, Danny Hernandez, Jackson Kernion, Kamal Ndousse, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark, Sam McCandlish, Chris Olah, and Jared Kaplan. A general language assistant as a laboratory for alignment, 2021.

Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, et al. Constitutional AI: Harmlessness from AI feedback, 2022.

Angelica Chen, Sadhika Malladi, Lily H. Zhang, Xinyi Chen, Qiuyi Zhang, Rajesh Ranganath, and Kyunghyun Cho. Preference learning algorithms do not learn preference rankings. In *Advances in Neural Information Processing Systems*, 2024a.

Zixiang Chen, Yihe Deng, Huizhuo Yuan, Kaixuan Ji, and Quanquan Gu. Self-play fine-tuning converts weak language models to strong language models. In *International Conference on Machine Learning*, 2024b.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems, 2021.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. QLoRA: Efficient finetuning of quantized LLMs. In *Advances in Neural Information Processing Systems*, 2023.

Hanze Dong, Wei Xiong, Deepanshu Goyal, Yihan Zhang, Winnie Chow, Rui Pan, Shizhe Diao, Jipeng Zhang, Kashun Shum, and Tong Zhang. RAFT: Reward rAnked FineTuning for generative foundation model alignment, 2023. TMLR 2023.

Ezieddin Elmahjub, Junaid Qadir, Abdullah Mushtaq, Rafay Naeem, Ibrahim Ghaznavi, and Waleed Iqbal. IslamicLegalBench: Evaluating LLMs knowledge and reasoning of islamic law across 1,200 years of islamic pluralist legal traditions, 2026.

Leo Gao, Jonathan Tow, Baber Abbasi, Stella Biderman, Sid Black, Anthony DiPofi, Charles Foster, Laurence Golding, Jeffrey Hsu, Alain Le Noac'h, Haonan Li, Kyle McDonell, Niklas Muennighoff, Chris Ociepa, Jason Phang, Laria Reynolds, Hailey Schoelkopf, Aviya Skowron, Lintang Sutawika, Eric Tang, Anish Thite, Ben Wang, Kevin Wang, and Andy Zou. A framework for few-shot language model evaluation. `https://github.com/EleutherAI/lm-evaluation-harness`, 2023. Zenodo, doi:10.5281/zenodo.5371628.

Caglar Gulcehre, Tom Le Paine, Srivatsan Srinivasan, Ksenia Konyushkova, Lotte Weerts, Abhishek Sharma, Aditya Siddhant, Alex Ahern, Miaosen Wang, Chenjie Gu, Wolfgang Macherey, Arnaud Doucet, Orhan Firat, and Nando de Freitas. Reinforced self-training (ReST) for language modeling, 2023.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *International Conference on Learning Representations*, 2021.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

M. Waleed Kadous and Benjamin Olsen. JaleesBench: Are AI assistants good spiritual company?, 2026. Companion paper. Code and results: `https://github.com/iaser-ai/jaleesbench`.

Abderraouf Lahmar, Md Easin Arafat, Zakarya Farou, and Mufti Mahmud. IslamTrust: A benchmark for LLMs alignment with islamic values. In *5th Muslims in ML Workshop (MusIML), NeurIPS 2025*, 2025. OpenReview:PBcv90iKFB.

Arka Pal, Deep Karkhanis, Samuel Dooley, Manley Roberts, Siddartha Naidu, and Colin White. Smaug: Fixing failure modes of preference optimisation with DPO-Positive, 2024.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, 2023.

Noam Razin, Sadhika Malladi, Adithya Bhaskar, Danqi Chen, Sanjeev Arora, and Boris Hanin. Unintentional unalignment: Likelihood displacement in direct preference optimization, 2024.

Avi Singh, John D. Co-Reyes, Rishabh Agarwal, Ankesh Anand, Piyush Patil, Xavier Garcia, Peter J. Liu, James Harrison, Jaehoon Lee, Kelvin Xu, Aaron Parisi, et al. Beyond human data: Scaling self-training for problem-solving with language models, 2024. TMLR 2024.

Fahim Tajwar, Anikait Singh, Archit Sharma, Rafael Rafailov, Jeff Schneider, Tengyang Xie, Stefano Ermon, Chelsea Finn, and Aviral Kumar. Preference fine-tuning of LLMs should leverage suboptimal, on-policy data. In *International Conference on Machine Learning*, 2024.

Lewis Tunstall, Edward Beeching, Nathan Lambert, Nazneen Rajani, Kashif Rasul, Younes Belkada, Shengyi Huang, Leandro von Werra, Clémentine Fourrier, Nathan Habib, Nathan Sarrazin, Omar Sanseviero, Alexander M. Rush, and Thomas Wolf. Zephyr: Direct distillation of LM alignment, 2023.

Weizhe Yuan, Richard Yuanzhe Pang, Kyunghyun Cho, Xian Li, Sainbayar Sukhbaatar, Jing Xu, and Jason Weston. Self-rewarding language models, 2024.

Jeffrey Zhou, Tianjian Lu, Swaroop Mishra, Siddhartha Brahma, Sujoy Basu, Yi Luan, Denny Zhou, and Le Hou. Instruction-following evaluation for large language models, 2023.